

HYDROVOLTA

HIGH-RECOVERY WATER TREATMENT

HYDROVOLTA
Water Desalination and Reuse

THE COMPLETE WATER TREATMENT SOLUTION

SalinBloc™

A containerised, plug-and-play desalination plant — combining the complete treatment train and the patented SonixED™ core in a single deployable unit.

Patent granted 2021 · EIC Accelerator €2.2M · Founded 2016 · Leuven, Belgium · Active in Morocco & Perth

THE PRODUCT

One container. The complete treatment plant.

SalinBloc™ is Hydrovolta's standardised, containerised treatment platform. It integrates the entire process train — intake, pre-treatment, the patented SonixED™ desalination core, and post-treatment — into a single ISO container, configured around each client's water chemistry and end-use requirement.

It arrives pre-assembled, connects to an existing borehole, and is operational in days — without civil works or specialist contractors. Every unit runs automatically, with full cloud telemetry from day one.



SalinBloc™ 20 ft container — ready for deployment

98%	2-5%	0.19-	0.5-2.5
WATER RECOVERY, SUITABLE APPS	RESIDUAL BRINE, SELECTED CASES	KWH / M ³ ENERGY, FROM	BAR OPERATING PRESSURE

SYSTEM SPECIFICATION

Format	20 ft & 40 ft ISO container units
Throughput	2-60 m ³ /h by format — see Formats & Scale-up
Feed range	Brackish groundwater, 1,500-20,000 ppm TDS
Operation	Fully automated · cloud telemetry, logging & remote support access
Power	Grid or solar-hybrid compatible — built for arid and off-grid sites
Scale pathway	Parallel unit docking for modular capacity expansion

FORMATS & SCALE-UP

Sized to your flow. Scaled when you need more.

SalinBloc™ is available in two containerised formats, with the project-engineered SonixED™ skid for industrial integration. Every format runs the same SonixED™ core, and units synchronise in parallel — so capacity grows step by step, without replacing what is already installed.



SALINBLOC™ · CONTAINERISED

20 ft Unit

2–30 m³/h

≈ 44–660 m³/day per unit

Pilots, decentralised supply, agricultural perimeters and remote communities.

Recovery up to 98% · 0.5–2.5 bar



SALINBLOC™ · CONTAINERISED

40 ft Unit

15–60 m³/h

≈ 330–1,320 m³/day per unit

Regional utilities, larger agricultural and industrial sites with higher demand.

Recovery up to 98% · 0.5–2.5 bar

Capacity is open-ended — units synchronise in parallel, so combined daily flow scales without an upper limit.

Daily figures are shown per unit at 22 operating hours; actual flow depends on feed-water chemistry and the target recovery rate.

MODULAR BY DESIGN

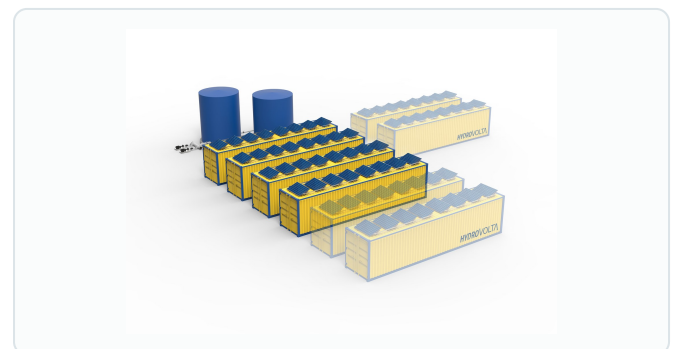
Start with one unit. Add capacity in parallel.

Every unit is scalable. Containers dock and synchronise to run in parallel, so flow is added in steps as demand grows — and for very high, utility- or industrial-scale volumes, project-engineered SonixED™ skids run in parallel trains.



One unit, operational in days

A single SalinBloc™ connects to an existing borehole and runs from day one — grid or solar-hybrid powered, fully automated.



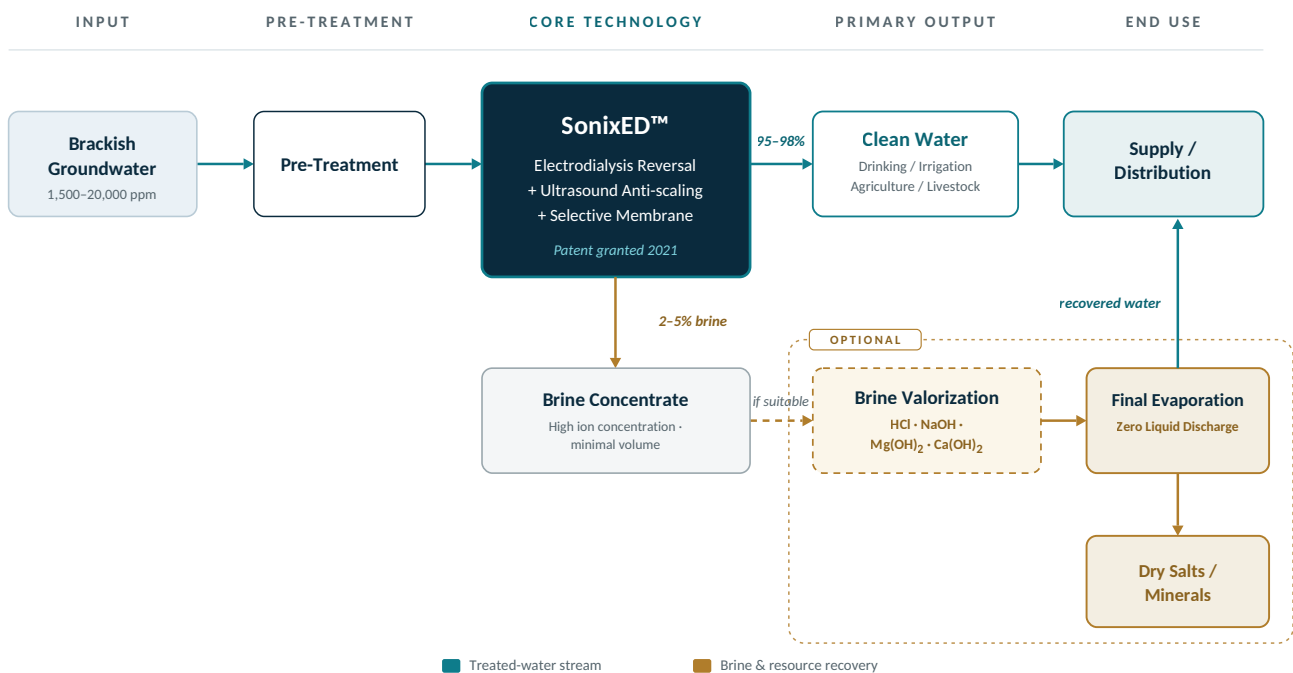
Synchronised units — step-by-step scale-up

Units synchronise to operate in parallel, adding capacity as demand grows. For full-scale flows, parallel SonixED™ skid trains deliver high throughput.

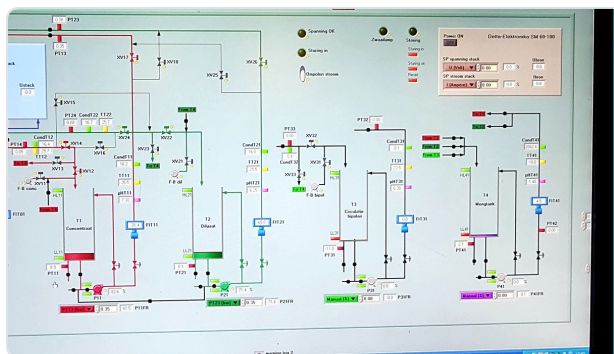
THE COMPLETE WATER CYCLE

From one brackish source to zero liquid discharge.

A single SalinBloc™ unit recovers up to 98% of the feed as clean water, then drives the small residual stream through optional valorization and final evaporation — recovering chemicals, additional water, and dry minerals. Nothing is discharged.



The story, end to end: brackish groundwater (1,500–20,000 ppm) is conditioned, then desalinated in the patented SonixED™ core to drinking, irrigation, or process quality. The 2–5% residual brine is concentrated; the optional recovery stages — valorization of acids, bases and mineral hydroxides, then final evaporation — engage only where the brine chemistry and project economics support them. When engaged, distilled water returns to supply and the remainder leaves as dry, handleable salts, achieving zero liquid discharge.



CLOUD MONITORING & TELEMETRY

Real-time visibility across every stage.

Flow, pressure, energy and water quality monitored live — with remote control, alerts, a client dashboard, and Hydrovolta support access, from intake to final discharge.

— CORE TECHNOLOGY — SONIXED™

Three integrated mechanisms. One validated platform.



SonixED™ integrates electro dialysis reversal, ultrasound-assisted scaling control, and a proprietary membrane coating into one automated platform. Each component addresses a specific failure mode of membrane treatment in the 1,500–20,000 ppm brackish range — together delivering results that individual technologies cannot achieve alone.

01 ⚡ Electro dialysis Reversal (EDR)

An electric potential drives dissolved ions through ion-exchange membranes — the water itself does not pass through. Periodic polarity reversal cleans the membranes in-situ, without chemical intervention.

- Stable output under variable feed TDS
- No chemical CIP dependency
- Lower pressure and energy than RO

02 📶 Ultrasound Integration — Patent 2021

Hydrovolta's granted patent places ultrasound transducers at the membrane surface. Acoustic cavitation continuously disrupts scale nucleation and prevents silica fouling — the primary failure mode that defeats conventional systems on hard and silica-laden water.

- Prevents calcium, magnesium & silica scale
- 50–90% fewer CIP cycles in validated contexts
- Extended membrane lifespan

03 🌊 Selective Membrane Coating

A proprietary coating enables monovalent ion selectivity — removing nitrate, sodium, chloride and fluoride while retaining beneficial calcium and magnesium. Output meets drinking and agricultural standards without full demineralisation.

- Selective nitrate removal — validated at Water Corporation Perth
- Mineral-balanced output, no remineralisation cost
- Ion targets configured per application

Developed over eight years through institutional research partnerships including VITO and CNR. Patent granted 2021. All performance figures are qualified by feedwater chemistry and operating conditions.

VALIDATED PERFORMANCE

Measured against conventional membrane treatment.

Independent comparison from the Germany 2023 industrial pilot — the same feed water (TDS 4,500 ppm, silica 85 mg/l, high hardness) treated over 90 days.

PARAMETER	SONIXED™	REVERSE OSMOSIS	STANDARD EDR
Water recovery	90–98%	50–75%	70–85%
Energy consumption	0.19–0.7 kWh/m ³	2.0–4.0 kWh/m ³	1.0–2.0 kWh/m ³
Chemical dosing	Minimal — no anti-scalant	Required	Anti-scalants needed
Operating pressure	0.5–2.5 bar	25–35 bar	1–2 bar
CIP cleaning frequency	0.6×/month	1×/month	1×/month
Treatment cost	\$0.30–0.40/m ³	\$0.60–1.00/m ³	\$0.40–0.80/m ³

Source: Hydrovolta Germany industrial pilot, Feb–May 2023. Figures qualified by feedwater chemistry and operating conditions.

WHY IT MATTERS TO THE OPERATOR

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Up to 98% recovery
Near-zero reject stream removes the brine-disposal barrier that blocks conventional plant expansion.
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No silica fouling
Ultrasound anti-scaling cuts cleaning cycles and extends equipment life on the hardest sources.
- 

Operational in days
Pre-assembled, borehole-ready, no civil works — deployed without specialist contractors.
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Low energy, low pressure
From 0.19 kWh/m³ at 0.5–2.5 bar — a fraction of the energy and pressure that reverse osmosis demands.
- 

Mineral-balanced water
Selective removal keeps calcium and magnesium — protecting soil, crop and process quality.
- 

Scales with demand
Start with one unit and dock further units in parallel as capacity requirements grow.

REFERENCE DEPLOYMENTS

Operating in the field. Validated before scale.

Hydrovolta has active equipment generating operational data, a contracted utility pilot, and signed pipeline across four continents — each labelled accurately by maturity.

ACTIVE PROJECT

Morocco — Saline Desalination + Brine Valorization

Seawater-intruded coastal aquifer at ~7,000 ppm TDS. SonixED™ desalination with integrated downstream brine valorization. Funded by the Flanders Climate Action Program. Primary integrated reference site.

UTILITY PILOT

Perth, Australia — Water Corporation

Selective nitrate and salinity removal across three public utility sites, supplying drinking, irrigation and livestock water. Cloud telemetry and regulatory monitoring active. Primary utility reference.

LOI SIGNED

Brazil — Remote Groundwater Remediation

Development-stage project addressing nitrate and salinity co-contamination, where high recovery and minimal brine disposal are central to the case.

MOU · PILOT IN DEVELOPMENT

Saudi Arabia — Selective Ion Recovery

Selective ion recovery from brine streams. Pilot design in progress.

DEPLOYMENT PATHWAY

OPTIONAL

STEP 01

Water analysis

Source chemistry reviewed, technical fit assessed. A clear GO / NO-GO before any hardware is specified.

STEP 02

Pilot design

SalinBloc™ configured for your source; operating case benchmarked and funding pathway identified.

STEP 03

On-site pilot

Commissioned with full monitoring; real operating data generated and brine potential evaluated.

STEP 04

Scale

Full installation engineered from pilot data — parallel units or engineered skid, only when data supports it.

Every project starts with the water.

Send your feed-water chemistry, target flow and the challenge you are solving. Our engineering team responds with an honest assessment of fit — typically within 48 hours.

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INSTITUTIONAL PARTNERS & BACKING



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